

OCR A-Level Computer Science (H446)
Component 01: Computer Systems
Section 1.1 Mock Exam

OCR Examination

Instructions to Candidates

- The total mark for this paper is 50.
- The marks for each question are shown in brackets [].
- Quality of extended response will be assessed in questions marked with an asterisk (*).
- This document consists of 4 pages.

Answer all questions

1. Define the term 'processor' and briefly describe its primary function in a computer system. [2 marks]
2. (a) State the meaning of the acronym 'CPU'. [1 mark]
(b) Explain the function of the Control Unit within a CPU. [2 marks]
3. The diagram below shows a simplified von Neumann architecture.

[Diagram showing CPU with Control Unit, ALU, Registers, connected to Memory and I/O]

- (a) Identify and explain the purpose of components A and B in the diagram. [4 marks]
- (b) Describe the function of the Address Bus and Data Bus in this architecture. [4 marks]
4. (a) Describe what is meant by the fetch-decode-execute cycle. [3 marks]
(b) Explain how pipelining can improve processor performance. [3 marks]
5. A computer manufacturer claims their new processor has a clock speed of 3.6 GHz.
 - (a) Explain what is meant by clock speed. [2 marks]
 - (b) Calculate how many cycles this processor can perform in one millisecond. Show your working. [2 marks]
 - (c) Discuss why clock speed alone is not always a reliable indicator of processor performance. [3 marks]
6. (a) Explain the difference between CISC and RISC processors. [4 marks]
(b) State one advantage and one disadvantage of RISC processors compared to CISC processors. [2 marks]
7. Describe the purpose of each of the following processor components:
 - (a) Arithmetic Logic Unit (ALU) [2 marks]

- (b) Registers [2 marks]
 - (c) Cache [2 marks]
8. (a) Explain what is meant by a multi-core processor. [2 marks]
- (b) Describe two benefits of multi-core processors. [2 marks]
9. *Discuss how the following factors affect the performance of a processor:
- Cache size and levels
 - Number of cores
 - Clock speed
 - Word length

In your answer, you should explain how each factor contributes to overall system performance and any trade-offs that might exist. [8 marks]

10. A user is choosing between two storage devices for their computer system:

Specification	Device A	Device B
Type	HDD	SSD
Capacity	2TB	500GB
Read Speed	150 MB/s	550 MB/s
Price	\$65	\$80

- (a) Compare the characteristics of HDD and SSD storage technologies. [4 marks]
- (b) Recommend which storage device would be most suitable for:
 - (i) A video editing workstation [2 marks]
 - (ii) A network file server with large storage requirements [2 marks]

Grade Boundaries

A*	A	B	C	D	E
45-50	40-44	35-39	30-34	25-29	20-24